

SINGLE-PHASE GAS FLOW IN MICROCHANNELS

2.1. Rarefaction and wall effects in microflows

2.1.1. Gas at the molecular level

2.1.1.1. Microscopic length scales

2.1.1.2. Binary intermolecular collisions in dilute simple gases

2.1.2. Continuum assumption and thermodynamic equilibrium

2.1.3. Rarefaction and Knudsen analogy

2.1.4. Wall effects

2.2. Gas flow regimes in microchannels

2.2.1. Ideal gas model

2.2.2. Continuum flow regime

2.2.2.1. Compressible Navier–Stokes equations

2.2.2.2. Classic boundary conditions

2.2.3. Slip flow regime

2.2.3.1. Continuum NS–QGD–QHD equations

2.2.3.2. First-order slip boundary conditions

2.2.3.3. Higher-order slip boundary conditions

2.2.3.4. Accommodation coefficients

2.2.4. Transition flow and free molecular flow

2.2.4.1. Burnett equations

2.2.4.2. DSMC method

2.2.4.3. Lattice Boltzmann method

2.3. Pressure-driven steady slip flows in microchannels

2.3.1. Plane flow between parallel plates

2.3.1.1. First-order solution

2.3.1.2. Second-order solutions

2.3.2. Gas flow in circular microtubes

2.3.2.1. First-order solution

2.3.2.2. Second-order solution

2.3.3. Gas flow in annular ducts

2.3.4. Gas Flow in rectangular microchannels

2.3.4.1. First-order solution

2.3.4.2. Second-order solution

2.3.5. Experimental data

2.3.5.1. Experimental setups for flow rate measurements

2.3.5.2. Flow rate data

2.3.5.3. Pressure data

2.3.5.4. Flow visualization

2.3.6. Entrance effects

2.4. Pulsed gas flows in microchannels

2.5. Thermally driven gas microflows and vacuum generation

2.5.1. Transpiration pumping

2.5.2. Accommodation pumping

2.6. Heat transfer in microchannels

2.6.1. Heat transfer in a plane microchannel

2.6.1.1. Heat transfer for a fully developed incompressible flow

2.6.1.2. Heat transfer for a developing compressible flow

2.6.2. Heat transfer in a circular microtube

2.6.3. Heat transfer in a rectangular microchannel

2.7. Future research needs

2.8. Solved examples

Example 2.1

Example 2.2

2.9. Problems

Problem 2.1

Problem 2.2

Problem 2.3

Problem 2.4

Problem 2.5

Problem 2.6

Appendix

Chapter 2-1

2.1. Rarefaction and wall effects in microflows

2.1.1. Gas at the molecular level

2.1.1.1. Microscopic length scales

2.1.1.2. Binary intermolecular collisions in dilute simple gases

2.1.2. Continuum assumption and thermodynamic equilibrium

2.1.3. Rarefaction and Knudsen analogy

2.1.4. Wall effects

Chapter 2

SINGLE-PHASE GAS FLOW IN MICROCHANNELS

As microfluidics concerns smaller sizes – the inner sizes of Micro-Electro-Mechanical-Systems (MEMS) – **new issues** have to be considered in order to accurately **model gas microflows**.

These issues are **mainly** due to

rarefaction effects,

which typically must be taken into account when

characteristic lengths are of the order of **1 μ m,**

under usual temperature and pressure conditions.

In this chapter, the role of **rarefaction** is explained and **its consequences** on the behavior of gas flows in microchannels are detailed.

The main theoretical and experimental results from the literature about **pressure-driven**, **steady** or **pulsed** gas microflows are summarized.

Heat transfer in microchannels and **thermally driven** gas microflows are also described. They are particularly interesting for **vacuum generation**, using microsystems **without** moving parts.

2.1. Rarefaction and wall effects in microflows

In microfluidics, theoretical knowledge for **gas flows** is currently **more advanced** than that for **liquid flows** (Colin, 2004). Concerning the gases, the issues are actually more clearly identified: the **main micro-effect** that results from shrinking down the devices size is **rarefaction**.

This allows us to exploit the strong, although not complete **analogy** between **microflows** and **low pressure flows** that has been extensively studied for more than 50 years and particularly for aerospace applications.

2.1.1. Gas at the molecular level

2.1.1.1. Microscopic length scales

Modeling gas microflows requires us to take into account, **several characteristic length scales**. At the molecular level, we may consider;

the mean molecular diameter d ,

the mean molecular spacing δ and,

the mean free path λ .

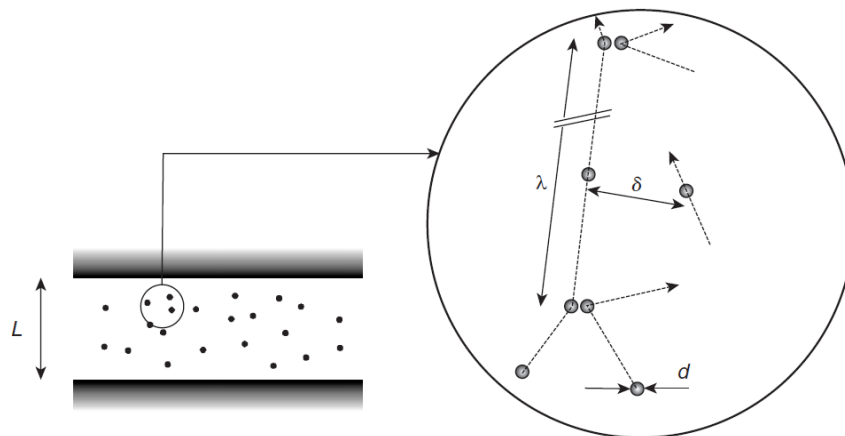


Fig. 2.1. Main characteristic length scales to take into account at the molecular level.

The **mean free path** is the **average distance traveled** by a molecule **between two consecutive collisions**, in a frame moving with the stream speed of the gas.

If we consider a **simple gas**, that is a gas consisting of a single chemical species with molecules having the same structure, then; the **mean free path** depends on their **mean diameter d** and on the **number density**:

$$n = \delta^{-3} \quad (2.1)$$

and the inverse **$1/n$** represents the **mean volume available for one molecule**.

$$\lambda = f(d, n) \quad \text{or} \quad \lambda = f(d, \delta) \quad \text{LK}$$

For example, a gas in standard conditions (i.e. for a temperature $T_0 = 273.15\text{K}$ and a pressure $p_0 = 1.013 \times 10^5 \text{ Pa}$), has about 27 million molecules in a cube of $1\mu\text{m}$ in width ($n_0 = 2.687 \times 10^{25}/\text{m}^3$ and $\delta_0 = 3.34 \text{ nm}$) and in the case of **air** for which **$d_0 \approx 0.42 \text{ nm}$** , the mean free path is **$\lambda_0 \approx 49 \text{ nm}$** .

When compared to the case of **liquid water**, the mean molecular diameter of each are nearly equal, but the mean molecular spacing is about 10 times **smaller** and the mean free path is **10^5 times smaller** than **air**!

2.1.1.2. Binary intermolecular collisions in dilute simple gases

dilute gases $\frac{d}{\delta} \ll 1$ (2.2)

Seyrek Gazlar

In that case, most of the intermolecular interactions are binary collisions.

Conversely, if this condition is not verified, the gas is said to be a dense gas.

The dilute gas approximation, along with the equipartition of energy principle, leads to the classic kinetic theory and the Boltzmann transport equation.

With dilute gas approximation;

Mean free path of the molecules $\lambda = \frac{\bar{c}'}{\nu} = \frac{\sqrt{8RT/\pi}}{\nu}$ (2.3)

$\bar{c}'$: Mean thermal velocity

ν : Collision rate

$$c' = c - u$$

c' : Thermal velocity c : Total velocity u : Local macroscopic velocity of flow

$\bar{c}' = \sqrt{8RT/\pi}$ Calculated from the Boltzmann equation

R : Specific gas constant T : Temperature

Model chosen for describing the elastic binary collision between two molecules enables to calculate:

ν , λ , as well as

collision cross-section σ_t ,

mean molecular diameter d ,

dynamic viscosity μ

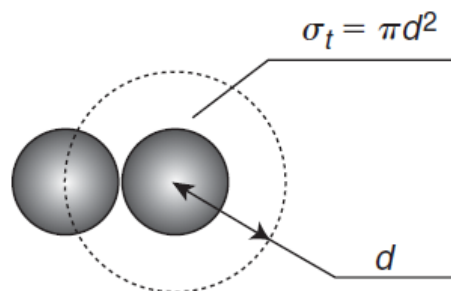


Fig. 2.2. Total collision cross-section in the collision plane.

A collision model generally requires the definition of the force $\mathbf{F}$ exerted between the two considered molecules.

This force is actually repulsive at short distances and weakly attractive at large distances.

Different approximated models are proposed to describe this force.

The more classic one (Bird, 1998) is the so-called;

inverse power law (IPL) model

or

point center of repulsion model.

It only takes into account the repulsive part of the force and assumes that

$$F = |\mathbf{F}| = \frac{\kappa}{r^\eta} \quad (2.5)$$

This derives from an intermolecular potential

$$\phi = \frac{\kappa}{(\eta - 1)r^{\eta-1}} \quad (2.6)$$

where κ , as well as the exponent η have a constant value and r denotes the distance between the two molecules centers.

Using the Chapman–Enskog theory, this model leads to a law of viscosity in the form

$$\mu = \mu_0 \left(\frac{T}{T_0} \right)^\omega \quad (2.7)$$

$$\omega = \frac{\eta + 3}{2(\eta - 1)} \quad (2.8)$$

The collision rate may be deduced as (Lengrand and Elizarova, 2004)

$$\nu = 4 \left(\frac{5}{2} - \omega \right)^{\omega - \frac{1}{2}} \Gamma \left(\frac{5}{2} - \omega \right) n \sigma_{t0} \left(\frac{T}{T_0} \right)^{\frac{1}{2} - \omega} \sqrt{\frac{RT}{\pi}} \quad (2.9)$$

$$\Gamma(j) = \int_0^\infty x^{j-1} \exp(-x) dx \quad \text{Euler, or Gamma function}$$

T_0 a reference temperature

σ_{t0} total collision cross-section calculated at reference temperature

From Eq. (2.3), we can see that the mean free path

$$\lambda = \frac{1}{\sqrt{2}(\frac{5}{2} - \omega)^{\omega - \frac{1}{2}} \Gamma(\frac{5}{2} - \omega) n \sigma_{t0}} \left(\frac{T}{T_0} \right)^{\omega - \frac{1}{2}} \quad (2.10)$$

Hard Sphere (HS) model → simplest collision model

The model assumes → 1) total collision cross-section σ_t is constant, and
2) viscosity is proportional to the square root of the temperature

Actually, the **HS** model may be described from the **IPL** with;

the exponent $\eta \rightarrow \infty$ and $\omega = 1/2$

Variable Hard Sphere (VHS) model → improved HS model

The VHS model may be considered as a HS model with a **diameter d** that is a function of the **relative velocity** between the **two colliding molecules**.

The Chapman–Enskog method leads to a viscosity

$$\mu_{VHS} = \frac{15 m \sqrt{\pi R T}}{8(\frac{5}{2} - \omega)^{\omega - \frac{1}{2}} \Gamma(\frac{9}{2} - \omega) \sigma_{t0}} \left(\frac{T}{T_0} \right)^{\omega - \frac{1}{2}} \quad (2.11)$$

m : Molecular mass

By eliminating σ_{t0} between Eqs. (2.9) or (2.10) and (2.11), and
By noting that $\Gamma(j + 1) = j\Gamma(j)$,

Collision rate:

$$\nu_{VHS} = \frac{\rho R T}{\mu} \frac{30}{(7 - 2\omega)(5 - 2\omega)} \quad (2.12)$$

Mean free path:

$$\lambda_{VHS} = \frac{\mu}{\rho \sqrt{2\pi R T}} \frac{2(7 - 2\omega)(5 - 2\omega)}{15} \quad (2.13)$$

Local density $\rho = mn$

Maxwell molecules (MM) model → in fact IPL model with $\eta=5$ and $\omega=1$

Actually, the HS and the MM models may be considered as the limits of the more realistic VHS model, since real molecules generally have a behavior which corresponds to an intermediate value $1/2 \leq \omega \leq 1$.

Another expression for mean free path

Karniadakis and Beskok (2002) or Nguyen and Wereley (2002) use the formula

$$\lambda_M = \sqrt{\pi/2} \mu / (\rho \sqrt{RT}) \quad \text{proposed by Maxwell in 1879 (Eq. (2.57)).}$$

Other binary collision models based on the IPL assumption

Variable soft sphere (VSS) model,

$$\lambda_{VSS} = \beta \lambda_{VHS} \quad \nu_{VSS} = \frac{\nu_{VHS}}{\beta} \quad (2.14)$$

$$\beta = 6\alpha / [(\alpha + 1)(\alpha + 2)] \quad 1 < \alpha < 2 \quad (\text{Bird, 1998})$$

Correction
Mean free path $\Rightarrow$ less than 3%.

For $\alpha = 1$ VSS model = VHS Model

Finally, some models can take into account the long-range attractive part of the force between two molecules by adding a uniform attractive potential (square-well model), or an IPL attractive component to the HS model.

Summary for IPL models

Table 2.1

Mean free path, dynamic viscosity, and collision rate for classic IPL collision models.

Model	$F = \frac{\kappa}{r^\eta}$ η	$\mu \propto T^\omega$ ω	$\nu = k_1 \frac{\rho RT}{\mu}$ k_1	$\lambda = k_2 \frac{\mu}{\rho \sqrt{RT}}$ k_2
HS	∞	$\frac{1}{2}$	$\frac{5}{4}$	$\frac{16}{5\sqrt{2\pi}} \approx 1.277$
VHS	η	$\omega = \frac{\eta + 3}{2(\eta - 1)}$	$\frac{30}{(7 - 2\omega)(5 - 2\omega)}$	$\frac{2(7 - 2\omega)(5 - 2\omega)}{15\sqrt{2\pi}}$
MM	5	1	2	$\sqrt{\frac{2}{\pi}} \approx 0.798$
VSS	η	$\omega = \frac{\eta + 3}{2(\eta - 1)}$	$\frac{5(\alpha + 1)(\alpha + 2)}{\alpha(7 - 2\omega)(5 - 2\omega)}$	$\frac{4\alpha(7 - 2\omega)(5 - 2\omega)}{5(\alpha + 1)(\alpha + 2)\sqrt{2\pi}}$

Mean free path λ is an **important parameter** for the simulation of gas microflows.

Summary for VHS or a VSS model

Table 2.2

Molecular weight, dynamic viscosity and mean molecular diameters under standard conditions ($p_0 = 1.013 \times 10^5$ Pa and $T_0 = 273.15$ K) estimated from a VHS or a VSS model (data from Bird (1998)) with corresponding values of the ratio δ_0/d .

pm picometre. $= 1 \times 10^{-12}$

Gas	$M \times 10^3$ (kg/mol)	$\mu_0 \times 10^7$ (N-s/m ²)	ω	d_{VHS} (pm)	α	d_{VSS} (pm)	δ_0/d
Sea level air	28.97	171.9	0.77	419	—	—	7.97
Ar	39.948	211.7	0.81	417	1.40	411	8.06
CH ₄	16.043	102.4	0.84	483	1.60	478	6.95
Cl ₂	70.905	123.3	1.01	698	—	—	4.78
CO	28.010	163.5	0.73	419	1.49	412	8.04
CO ₂	44.010	138.0	0.93	562	1.61	554	5.98
H ₂	2.0159	84.5	0.67	292	1.35	288	11.51
HCl	36.461	132.8	1.00	576	1.59	559	5.88
He	4.0026	186.5	0.66	233	1.26	230	14.42
Kr	83.80	232.8	0.80	476	1.32	470	7.06
N ₂	28.013	165.6	0.74	417	1.36	411	8.06
N ₂ O	44.013	135.1	0.94	571	—	—	5.85
Ne	20.180	297.5	0.66	277	1.31	272	12.16
NH ₃	17.031	92.3	1.10	594	—	—	5.62
NO	30.006	177.4	0.79	420	—	—	7.95
O ₂	31.999	191.9	0.77	407	1.40	401	8.26
SO ₂	64.065	116.4	1.05	716	—	—	4.66
Xe	131.29	210.7	0.85	574	1.44	565	5.86

2.1.2. Continuum assumption and thermodynamic equilibrium

Continuum assumption **erase** → Molecular discontinuities
by averaging the microscopic quantities on a small sampling volume

All macroscopic quantities **vary** → continuously from point to point
(density ρ , velocity $\mathbf{u}$, pressure p , temperature T , etc.)

For example, air flow in a duct

{ macroscopic velocity varies from 0 to $1 \frac{\text{m}}{\text{s}}$ and direction is parallel to the axis
However,
velocity of a molecule is of the order of 1 km/s and may take any direction

Similar considerations also concern the other mechanical and thermodynamic quantities.

Continuum assumption:

Microscopic fluctuations
should not generate
significant fluctuations of the averaged quantities

Consequently,

size of a representative sampling volume
must be **large enough** to erase
microscopic fluctuations,
but it must also be **small enough** to point out
macroscopic variations

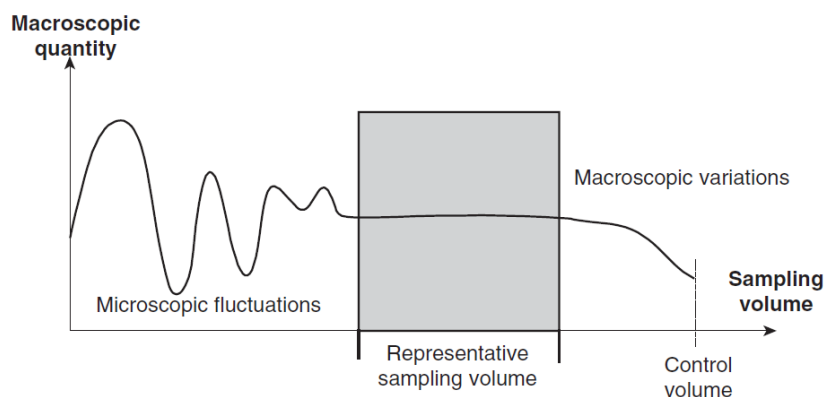


Fig. 2.3. The existence of a representative **sampling volume** (shaded area) is necessary for the continuum assumption to be valid.

If the shaded area on Fig. 2.3 does not exist,
sampling volume is not representative, and
continuum assumption is not valid.

Karniadakis and Beskok, 2002

↓

Sampling a volume containing 10,000 molecules
leads to 1% statistical fluctuations in the macroscopic quantities

Such a fluctuation level needs a sampling volume which
characteristic length l_{sv} verifies $l_{sv}/\delta = 10^{4/3} \approx 22$.
Consequently, the control volume must have a much higher characteristic length
 L , that is

$$\frac{L}{\delta} \gg 10^{4/3} \quad (2.16)$$

In this case statistical fluctuations can be neglected.

For example, for air at standard conditions,
the value of l_{sv} corresponding to 1% statistical fluctuations is 72 nm.
This is comparable to the value of the mean free path $\lambda_0 = 49$ nm.

Moreover, the continuum approach requires that the sampling volume is in
thermodynamic equilibrium.

For the thermodynamic equilibrium to be respected, the frequency of the
intermolecular collisions inside the sampling volume must be high enough.

$$\nu \rightarrow \text{high enough}$$

This implies that the mean free path $\lambda = \bar{c}'/\nu$ must be small compared with the
characteristic length l_{sv} of the sampling volume,
itself being small compared with the

characteristic length L of the control volume.

$$(\lambda < l_{sv} < L \quad \text{LK})$$

As a consequence, the thermodynamic equilibrium requires that

$$\frac{\lambda}{L} \ll 1 \quad (2.17)$$

Definition:

Knudsen number

$$\text{Kn} = \frac{\lambda}{L} \quad (2.18)$$

Knudsen number plays a **very important role** in **gaseous microflows**

If λ is obtained from an **IPL** collision model,

$$\lambda = k_2 \mu / \rho \sqrt{RT} \quad (2.19)$$

with k_2 given by **Table 2.1**

Knudsen number can be related to the Reynolds number and the Mach number by the relationship

$$\text{Re} = \frac{\rho u L}{\mu} \quad \text{Ma} = \frac{u}{a} \quad \text{Kn} = k_2 \sqrt{\gamma} \frac{\text{Ma}}{\text{Re}} \quad (2.22)$$

Here, γ is the ratio of the specific heats of the gas and a , the local speed of sound. For an ideal gas (Anderson, 1990),

$$a = \sqrt{\gamma RT}$$

If **Ma > 0.3**, **compressibility** effects have to be taken into account.

Indicative Limit Values (proposed by Bird (Bird, 1998))

Dense Gas	Negligible Statistical Fluctuations	Thermodynamic Equilibrium
$\frac{d}{\delta} \ll 1$	$\frac{L}{\delta} \gg 10^{4/3}$	$\frac{\lambda}{L} \ll 1$
$\delta/d=7$	$L/\delta=100$	$\lambda/L=0.1$

Fig. 2.4. is drawn by taking into account the Indicative Limit Values. In this figure, the mean free path has been estimated with an HS model, for which Eq. (2.10) can be written using Eq. (2.4) and $\omega=1/2$, in the form

$$\lambda_{HS} = \frac{\delta^3}{\sqrt{2}\pi d^2} \quad (2.23)$$

Thermodynamic equilibrium limit $\lambda/L=0.1$ is a straight line with a slope equal to -3 in the **log-log** plot of Fig. 2.4.

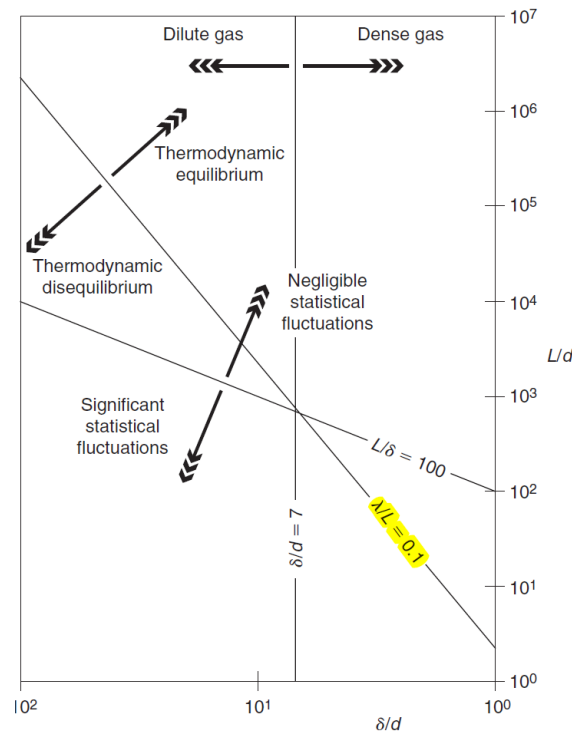


Fig. 2.4. Limits of the main approximations for the modeling of gas microflows (from Bird (1998)). **log-log** plot

2.1.3. Rarefaction and Knudsen analogy

It is now clear that the similitude between low pressure and confined flows is not complete, since;

the Knudsen number is not the only parameter to take into account (cf. Fig. 2.4.).

Thermodynamic equilibrium is another parameter.

However, it is convenient to differentiate the flow regimes in function of Kn , and the following classification, although tinged with empiricism, is usually accepted:

- For $Kn < 10^{-3}$, the flow is a continuum flow (C)

The flow is accurately modeled by the compressible Navier–Stokes equations with classical no-slip boundary conditions.

- For $10^{-3} < Kn < 10^{-1}$, the flow is a slip flow (S)

The Navier–Stokes equations remain applicable, provided;

a velocity slip and a temperature jump

are taken into account at the walls. These new boundary conditions point out that rarefaction effects become sensitive at the wall first.

- For $10^{-1} < Kn < 10$, the flow is a transition flow (T)

The continuum approach of the Navier–Stokes equations is no longer valid. However, the intermolecular collisions are not yet negligible and should be taken into account.

- For $Kn > 10$, the flow is a free molecular flow (M)

The occurrence of intermolecular collisions is negligible compared with the one of collisions between the gas molecules and the walls.

These different regimes will be detailed in the following sections.

Their limits are only indicative and could vary from one case to another, partly because the choice of the

characteristic length L

is rarely unique.

For flows in channels, L is generally the
hydraulic diameter
or the
depth of the channel

In **complex geometrical configurations**, it is generally preferable to define L from

local gradients (e.g. of the density ρ : $L=1/|\nabla \rho/\rho|$)

rather than from

simple geometrical considerations (Gad-el-Hak, 1999);

the **Knudsen number** based on this characteristic length is the so-called
local rarefaction number (Lengrand and Elizarova, 2004).

Figure 2.5 locates these different regimes for **air** in standard conditions. For comparison, the cases of **helium** and **sulphur dioxide** are also represented.

The **shaded area** roughly represents the domain of validity for classic gas flow models (C).

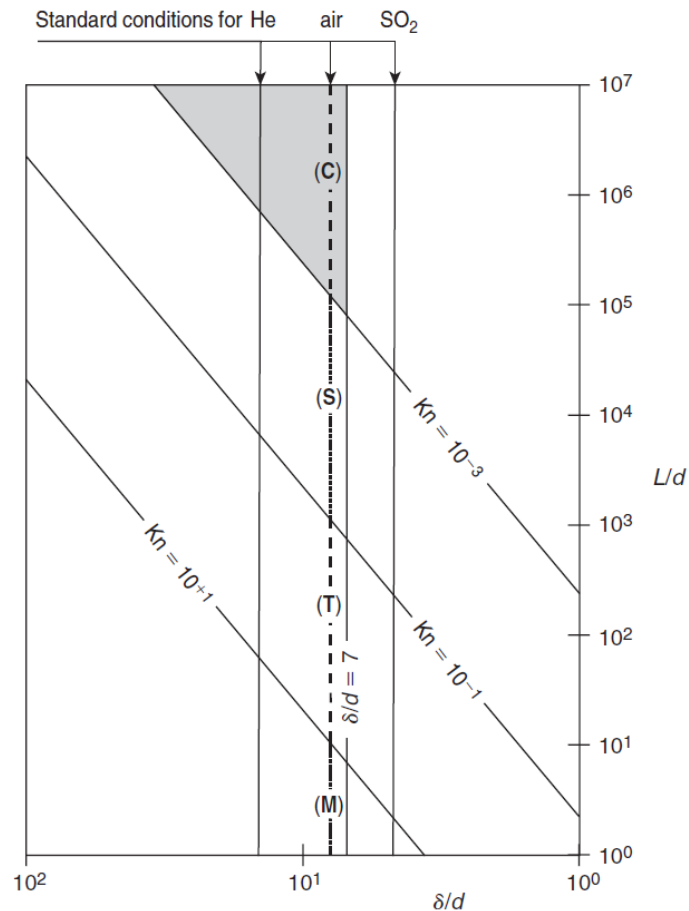


Fig. 2.5. Gas flow regimes as a function of the Knudsen number.

The relationship with the characteristic length L expressed in micrometers (μm LK) is illustrated in Fig. 2.6, which shows the typical ranges covered by fluidic microsystems described in the literature.

Typically, most of the microsystems which use gases work;

in the slip flow regime,

or

in the early transition regime.

In simple configurations, such flows can be analytically or semi-analytically modeled.

The core of the transition regime relates to more specific flows that involve lengths

under 100 nanometers,

as it is the case for a Couette flow between a hard disk and a read-write head. In that regime, the only theoretical models are molecular models that require numerical simulations.

Finally, under the effect of **both low pressures** and **small dimensions**, more rarefied regimes can occur, notably inside microsystems dedicated to **vacuum generation**.

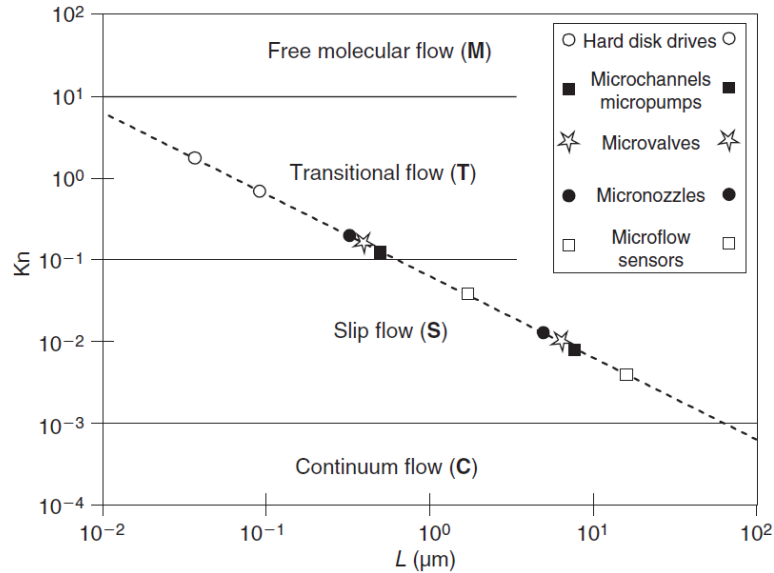


Fig. 2.6. Gas flow regimes for usual microsystems (from Karniadakis and Beskok (2002)). Assume **air** flow with a mean free path equal to $0.05 \mu\text{m}$ in a microchannel. What would be the upper and lower **limits of characteristic length** of microchannel in different flow regimes?

$$Kn = \frac{\lambda}{L} \quad L = \frac{\lambda}{Kn} \quad Kn < \rightarrow L >$$

- For **$Kn < 10^{-3}$** , the flow is a **continuum flow (C)**

$$\infty > L > \frac{0.05}{10^{-3}} \mu\text{m} \quad \infty > L > 5 \times 10 \mu\text{m}$$

$$50 \mu\text{m} < L < \infty$$

- For **$10^{-3} < Kn < 10^{-1}$** , the flow is a **slip flow (S)**

$$\frac{0.05}{10^{-3}} > L > \frac{0.05}{10^{-1}} \mu\text{m} \quad 5 \times 10 > L > 5 \times 10^{-1} \mu\text{m}$$

$$0.5 \mu\text{m} < L < 50$$

- For **$10^{-1} < Kn < 10$** , the flow is a **transition flow (T)**

$$\frac{0.05}{10^{-1}} > L > \frac{0.05}{10} \mu\text{m} \quad 5 \times 10^{-1} > L > 5 \times 10^{-3} \mu\text{m}$$

$$0.005 \mu\text{m} < L < 0.5$$

- For **$Kn > 10$** , the flow is a **free molecular flow (M)**

$$\frac{0.05}{10} > L > 0 \mu\text{m} \quad 5 \times 10^{-3} > L > 0 \mu\text{m}$$

$$0 \mu\text{m} < L < 0.005$$

2.1.4. Wall effects

We have seen that initial deviations from the classic continuum models appear when we shrink down the characteristic length L of the flow with respect to the boundary $\overbrace{\text{conditions}}^{\text{sil-LK}}$.

It is logical that the first sources of thermodynamic disequilibria appear on the boundary because

at the wall

there are fewer interactions between the gas molecules than in the core of the flow.

Actually, the walls play a crucial role in gas microflows.

Shrinking system sizes results in an increase of surface over volume effects.